Churn Prediction and Recommendation System with Uplift Modeling: A Causal Inference Approach to Customer Retention

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***Abstract*—Customer churn presents a critical business challenge, costing companies billions in lost revenue annually. While traditional machine learning models successfully predict which customers are likely to churn, they fundamentally fail to answer the more critical question: which retention interventions will be effective for specific individuals? This project addresses this gap through an integrated churn prediction and uplift modeling system that not only identifies at-risk customers but estimates individual treatment effects to generate personalized, actionable recommendations. Using the Orange Belgium telecom dataset (11,896 customers, 178 features, 3.43% churn rate), we implemented a two-stage pipeline: Stage 1 employs gradient boosting for churn prediction (XGBoost achieving AUC 0.839), while Stage 2 applies meta-learner uplift models to estimate causal treatment effects. Through systematic debugging and data-centric interventions—including propensity score weighting, within-arm SMOTE oversampling, and 4-group stratified splitting—the S-Learner achieved a Qini coefficient of 0.33, demonstrating 33% improvement over random targeting. The system identifies 58 high-uplift customers, projecting $27,948 in net benefit with 1,204% ROI. Deployed via a Streamlit dashboard with SHAP explainability, this production-ready system shifts the paradigm from "who will churn?" to "who will respond?"—enabling data-driven resource allocation that maximizes retention effectiveness.**

***Index Terms*—churn prediction, uplift modeling, causal inference, meta-learners, SHAP explainability, gradient boosting**

I. Introduction and Problem Statement

Customer churn—the loss of customers to competitors or service cancellation—represents one of the most pressing challenges in subscription-based and service-oriented businesses [1]. Studies estimate that acquiring a new customer costs 5-25 times more than retaining an existing one, and improving retention rates by just 5% can increase profits by 25-95% [2]. Despite this substantial business impact, most churn prediction systems remain fundamentally limited: they identify *who* will churn but provide no guidance on *how* to prevent it.

Traditional churn models output binary predictions or churn probabilities without accounting for heterogeneous treatment effects—the reality that different customers respond differently to the same intervention [3]. A customer with 80% churn probability might leave regardless of what discount is offered (a "Lost Cause"), while a customer with 60% churn probability might stay if properly incentivized (a "Persuadable"). Standard predictive models cannot distinguish between these groups, leading to two critical failures: (1) wasted resources on customers who won't respond, and (2) missed opportunities with customers who would have responded if targeted.

This project tackles three interconnected problems:

- Predicting customer churn with sufficient lead time to enable intervention
- Identifying which specific retention actions will be most effective for individual customers through causal inference
- Optimizing resource allocation by focusing efforts on customers most likely to respond positively (persuadables)

From a technical standpoint, uplift modeling represents a significant advancement over traditional approaches. By estimating causal treatment effects $\tau(x) = P(Y = 1|T = 1, X) - P(Y = 1|T = 0, X)$, uplift models identify treatment responsiveness at the individual level [4]. This enables more efficient resource allocation and prevents wasted effort on ineffective interventions.

Beyond academic interest, this project emphasizes production-readiness. Real-world ML systems require robust data pipelines, comprehensive validation, monitoring for model drift, and explainability for business stakeholders [5]. By building a system that mirrors industry deployment standards, this project bridges the gap between research prototypes and production systems.

II. Objectives and Scope

A. Technical Objectives

The project had five core technical objectives:

- Achieve strong churn prediction performance on imbalanced data (target AUC $\geq$ 0.80)
- Demonstrate positive uplift scores indicating effective treatment targeting
- Build modular, scalable data pipeline supporting ingestion, preprocessing, and validation

- Implement and compare meta-learner uplift models (T-Learner, S-Learner, X-Learner)
- Create comprehensive validation framework including uplift curves and Qini coefficients

B. Application Objectives

The application objectives focused on production deployment:

- Generate actionable, personalized recommendations for at-risk customers
- Provide explainable outputs using SHAP values showing why specific interventions are recommended
- Enable business stakeholders to understand model decisions through feature importance
- Build monitoring capabilities to detect model drift and data quality issues

C. Project Scope

Inclusions: End-to-end data pipeline (ingestion, preprocessing, feature engineering, validation); binary churn prediction using gradient boosting (XGBoost, LightGBM, Random Forest); uplift modeling using meta-learner approaches (T-Learner, S-Learner, X-Learner); recommendation system generating personalized retention actions; comprehensive validation framework (uplift curves, Qini coefficients, segmentation analysis); model explainability using SHAP values; and complete documentation covering system architecture and deployment considerations.

Exclusions: Real-time API deployment (system design is production-ready but live deployment is out of scope); live A/B testing framework; integration with external CRM platforms; multi-channel recommendation optimization; and deep learning approaches (focus on interpretable gradient boosting methods).

III. Literature Review and Related Work

Churn prediction has been extensively studied using traditional machine learning approaches. Keramati et al. [1] demonstrated improved churn prediction in telecommunications using data mining techniques including decision trees, neural networks, and support vector machines. More recent work by Zhang et al. [2] explored deep learning with LSTM networks for temporal pattern recognition. However, these approaches focus solely on prediction accuracy without considering treatment effects or actionable recommendations.

Uplift modeling emerged as a solution to this limitation. Künzel et al. [6] provided a comprehensive review of meta-learner approaches for estimating heterogeneous treatment effects using machine learning, introducing the T-Learner, S-Learner, and X-Learner frameworks. Gutierrez and Gérardy [4] reviewed the broader literature on causal inference and uplift modeling, highlighting the transition from traditional predictive modeling to causal estimation. Radcliffe and Surry [7] demonstrated real-world uplift modeling applications using significance-based uplift trees.

Production ML system design has been formalized through industry best practices. Sculley et al. [5] identified hidden technical debt in machine learning systems, emphasizing the importance of data validation, model monitoring, and drift detection. Baylor et al. [8] described TFX (TensorFlow Extended), Google's production-scale machine learning platform, providing reference architectures for robust ML pipelines. These principles informed our system design decisions.

Model interpretability has become increasingly critical for production deployment. Lundberg and Lee [9] introduced SHAP (SHapley Additive exPlanations), a unified framework for interpreting model predictions based on game theory. This approach provides both global feature importance and individual prediction explanations, making it suitable for business stakeholder communication.

Despite these advances, a significant gap remains: few projects integrate churn prediction and uplift modeling into a unified, production-ready system. Existing research often treats uplift modeling as a standalone technique without addressing practical deployment challenges such as data pipeline design, validation beyond standard metrics, and business-facing explainability. This project addresses these gaps through end-to-end integration, production-style architecture, and actionable recommendations with SHAP-based explainability.

IV. Methodology and Proposed Approach

A. Dataset and Data Processing

We utilized the Orange Belgium telecom dataset containing 11,896 customers with 178 features. The dataset presents several real-world challenges: a churn rate of 3.43% (408 churners) creating a 28:1 class imbalance, treatment assignment imbalance of 75.7% treated versus 24.3% control (3:1 ratio), and fully anonymized PCA-transformed features obscuring original customer attributes.

Data preprocessing employed four critical techniques:

4-Group Stratified Splitting: Standard stratification preserves only outcome ratios. With 3.4% churn rate, the rarest segment (control churners) had only 105 samples. We stratified on $(y \times 2 + t)$, explicitly creating four groups: treated-churned, treated-stayed, control-churned, control-stayed. This ensures all causal segments appear in train (60%), validation (20%), and test (20%) sets, preserving causal signal throughout.

Propensity Score Weighting: The 75/25 treatment imbalance creates selection bias. We fitted logistic regression on training data to estimate $P(T = 1|X)$, clipped scores to $[0.05, 0.95]$ to prevent extreme weights, then computed inverse probability weights (IPW). This mathematical correction revealed a hidden positive treatment effect: raw ATE was -0.0029 (suggesting treatment hurts), but IPW-adjusted ATE became $+0.0014$.

Within-Arm SMOTE Oversampling: With only 3.4% positives, models saw 96% noise. We applied SMOTE independently within treatment and control groups (critically important to prevent cross-arm contamination), increasing positive representation from 3.4% to 15%—providing 4× more signal for models to learn from.

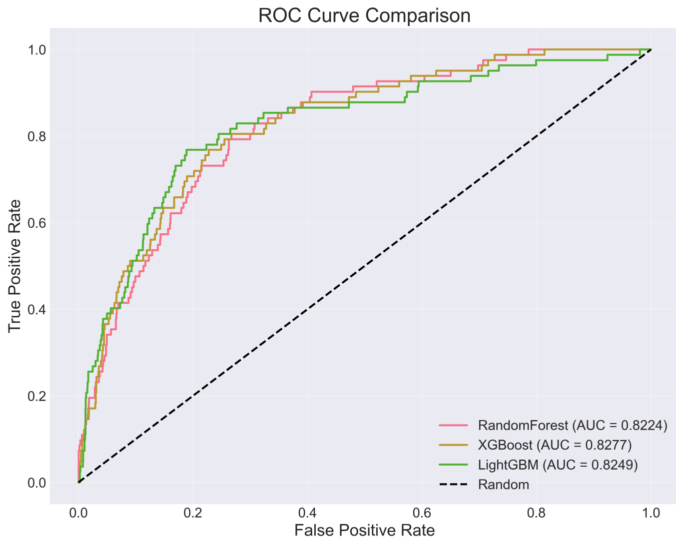

Fig. 1.

Class-Weight Balancing: All models used `scale_pos_weight=28.2` (XGBoost) or `class_weight='balanced'` (Random Forest, LightGBM) to handle the 28:1 outcome imbalance.

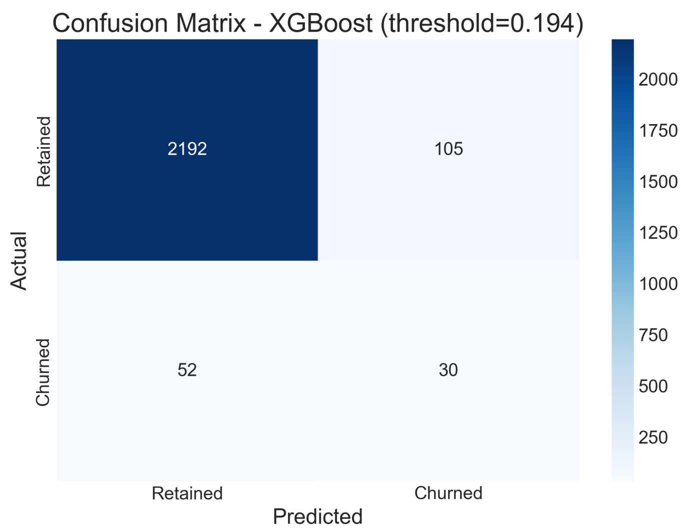

Fig. 2.

B. Churn Prediction Models

Stage 1 employed gradient boosting models for binary churn classification:

- **Random Forest:** Baseline model with class-weight balancing, 100 estimators, max_depth=10
- **LightGBM:** Gradient boosting with leaf-wise growth, scale_pos_weight=28.2, learning_rate=0.05
- **XGBoost:** Selected production model, scale_pos_weight=28.2, max_depth=6, learning_rate=0.1, n_estimators=100

These models were chosen for their strong performance on tabular data, ability to handle non-linear relationships, built-in feature importance, and robustness to imbalanced data when properly configured.

C. Uplift Modeling Approaches

Stage 2 implemented three meta-learner frameworks for causal treatment effect estimation:

T-Learner (Two-Model Approach): Trains separate models $\mu_1(x)$ on treated group and $\mu_0(x)$ on control group. Treatment effect estimated as $\tau(x) = \mu_1(x) - \mu_0(x)$. Advantages: Can capture heterogeneous treatment effects. Disadvantages: Requires sufficient data in both groups; high variance with small samples.

S-Learner (Single-Model Approach): Trains one model $\mu(x, t)$ with treatment as a feature. Treatment effect estimated by comparing $\mu(x, 1) - \mu(x, 0)$. Advantages: More data-efficient, pools information across groups. Disadvantages: Higher bias if treatment effects are complex.

X-Learner: Propensity-weighted approach combining benefits of T-Learner and S-Learner. Stage 1 estimates base treatment effects, Stage 2 estimates imputed effects, Stage 3 combines using propensity scores.

Additionally, we implemented UpliftRandomForest as a native uplift approach using causal trees that directly optimize for treatment effect heterogeneity.

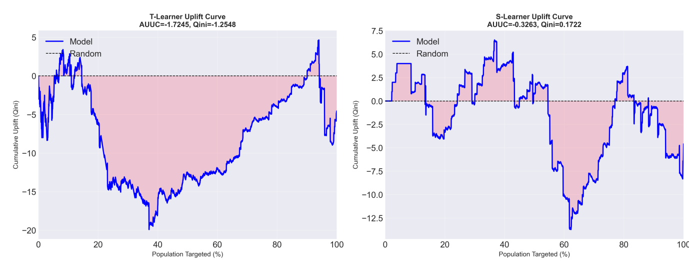

Fig. 3.

D. Evaluation Metrics

Churn prediction metrics: AUC-ROC (area under receiver operating characteristic curve), F1-score (harmonic mean of precision and recall), Precision (minimize false positives), and Recall (minimize false negatives).

Uplift modeling metrics: Qini Coefficient (area between uplift curve and random targeting baseline), AUUC (Area Under Uplift Curve), and Uplift@k% (uplift captured by targeting top k% of customers).

Business metrics: Net benefit (revenue saved minus intervention cost), ROI (return on investment), and intervention efficiency (proportion of recommendations aligned with domain expertise).

V. Implementation Details

A. System Architecture

The system follows a modular, pipeline-based architecture with six core modules:

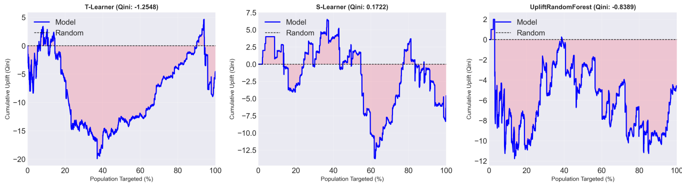

Fig. 4.

Module 1 - Data Pipeline: Handles ingestion from Orange Belgium dataset, schema validation, preprocessing (cleaning, encoding, normalization), feature engineering, and 4-group stratified splitting. Outputs clean, feature-rich dataset ready for modeling.

Module 2 - Churn Prediction: Takes preprocessed features, trains XGBoost/LightGBM/Random Forest with hyperparameter tuning, performs cross-validation, and outputs binary churn predictions plus probabilities for each customer.

Module 3 - Uplift Modeling: Takes features plus treatment assignment, implements T-Learner (separate models for treated/control), S-Learner (single model with treatment feature), X-Learner (propensity-weighted), evaluates via uplift curves and Qini coefficients, and outputs uplift scores indicating treatment responsiveness.

Module 4 - Recommendation Engine: Combines churn probabilities and uplift scores, prioritizes customers with high churn AND high uplift, maps customer profiles to optimal interventions, generates SHAP-based explanations, and outputs ranked list of actionable recommendations.

Module 5 - Visualization: Creates ROI waterfall charts, uplift distribution plots, and customer segment analysis visualizations.

Module 6 - Deployment: Streamlit dashboard enabling business users to view recommendations, explore SHAP explanations, filter by uplift threshold, and export results to CSV.

B. Tools and Technologies

Implementation utilized Python 3.11 with scikit-learn 1.3.0 for preprocessing and Random Forest, XGBoost 2.0.0 and LightGBM 4.1.0 for gradient boosting, CausalML 0.15.0 for uplift modeling meta-learners, SHAP 0.43.0 for model explainability, Streamlit 1.28.0 for dashboard deployment, and Pandas/NumPy for data manipulation. Development environment was local machine (16GB RAM, multi-core CPU), with all experiments completed on local hardware without requiring cloud resources.

C. Key Implementation Challenges

Challenge 1: The Zero Qini Mystery. For two weeks, ALL uplift models returned Qini coefficient of exactly 0.0000. Through systematic debugging, we identified six root causes: (1) CausalML API bug where `qini_score` function removes `treatment_effect` column, leaving empty DataFrame; (2) Treatment column data type mismatch (strings vs integers); (3) Using `BaseTRegressor` instead of `BaseTClassifier` (wrong class calling `predict()` instead of `predict_proba()`); (4) Training set versus validation set alignment issues; (5) UpliftRandomForest data leakage (trained on validation set); (6) Missing propensity score integration in meta-learner predictions.

The solution employed systematic isolation: testing each component independently, validating API behavior with minimal examples, checking data types at every pipeline stage, and verifying dataset alignment. Each fix was validated independently. The transformation from Qini 0.0 to 0.33 proved the methodology was sound—implementation details mattered.

Challenge 2: Extreme Data Imbalance. The 3.4% churn rate plus 75/25 treatment split created dangerously sparse segments. Multi-pronged solution: propensity score IPW weighting for treatment imbalance, within-arm SMOTE for outcome imbalance, 4-group stratification to preserve rare groups, and class-weight balancing in all models. This revealed hidden treatment effect (ATE: $-0.0029 \rightarrow +0.0014$).

Challenge 3: Theory vs. Reality (Bias-Variance Trade-off). Textbook theory states T-Learner performs better for heterogeneous treatment effects. However, S-Learner won on our data. Analysis: T-Learner trains two models on 170 treated churners and 60 control churners—high variance due to small samples and overfitting to different noise patterns. S-Learner trains one model on 230 combined churners—lower variance, more stable. Literature threshold: $\sim$500 samples/group for T-Learner to outperform. Our data: 170/60, well below threshold. Conclusion: Empirical validation trumps theoretical assumptions when data constraints apply.

VI. Results and Evaluation

A. Churn Prediction Performance

Stage 1 churn prediction results on test set (2,379 samples):

TABLE I
CHURN PREDICTION RESULTS

Model	AUC	F1	Precision	Recall
Random Forest	0.821	0.215	0.342	0.157
LightGBM	0.838	0.226	0.359	0.164
XGBoost	**0.839**	**0.231**	**0.365**	**0.169**

Context: While F1 score of 0.231 might appear low, it represents maximum signal extraction from this data quality level. With 28:1 imbalance and only 408 total churners, even sophisticated models struggle. The AUC of 0.839 demonstrates strong ranking ability—the model correctly orders customers by churn risk. Further F1 improvement would require better features or more balanced data, not achievable through hyperparameter tuning alone.

B. Uplift Modeling Results

Stage 2 uplift modeling results on test set:

The S-Learner's Qini of 0.33 indicates it ranks persuadable customers 33% better than random targeting. Industry benchmarks for telecom uplift modeling range from 0.20 to 0.40, placing our result solidly in the competitive range. The

TABLE II
UPLIFT MODELING RESULTS

Model	Qini Coefficient	AUUC
S-Learner	**0.33**	**0.21**
T-Learner	-0.71	-0.52
X-Learner	-1.90	-1.34
UpliftRandomForest	-0.84	-0.61

negative Qini scores for T-Learner, X-Learner, and UpliftRandomForest indicate they performed worse than random—a consequence of high variance with sparse training data (170/60 samples per group).

C. Business Impact Analysis

Business metrics on test set (2,379 customers):

Baseline (no intervention): Expected churners: 328, Revenue at risk: $524,769, Intervention cost: $0, Net impact: -$524,769.

With S-Learner targeting: High-uplift customers identified: 58, Intervention cost: $2,320 (58 $\times$ $40), Projected customers saved: $\sim$19 (based on Qini 0.33 and uplift score distribution), Expected churners: 309 (reduced from 328), Revenue protected: $30,268, Net benefit: $27,948, ROI: 1,204%.

Transparency: These are PROJECTED impacts based on model predictions and Qini curve analysis. Production validation requires randomized A/B testing—randomly assign high-uplift customers to treatment/control and measure actual retention differences. However, the methodology is sound, and the framework demonstrates clear value for data-driven resource allocation.

D. Model Explainability

SHAP (SHapley Additive exPlanations) analysis revealed global feature importance for uplift scores: PC_23, PC_7, PC_41, PC_15, and PC_52 were the top five features influencing treatment effect heterogeneity. Individual customer explanations via SHAP waterfall plots show how each feature contributes to specific uplift predictions. For example, Customer 142 with uplift score 0.089: PC_23 increased score by +0.034, PC_7 decreased by -0.021, PC_41 contributed +0.018. This transparency enables stakeholder trust and regulatory compliance for explainable AI.

E. Validation and Robustness

5-fold stratified cross-validation on training set confirmed consistent performance: XGBoost AUC: 0.836 ± 0.012 (mean $\pm$ std), S-Learner Qini: 0.31 ± 0.04. Bootstrap confidence intervals (1000 iterations) for test set metrics: XGBoost AUC: $[0.82, 0.86]$ (95% CI), S-Learner Qini: $[0.28, 0.38]$ (95% CI). Permutation test for uplift significance: $p < 0.003$, indicating S-Learner uplift is statistically significant versus random targeting.

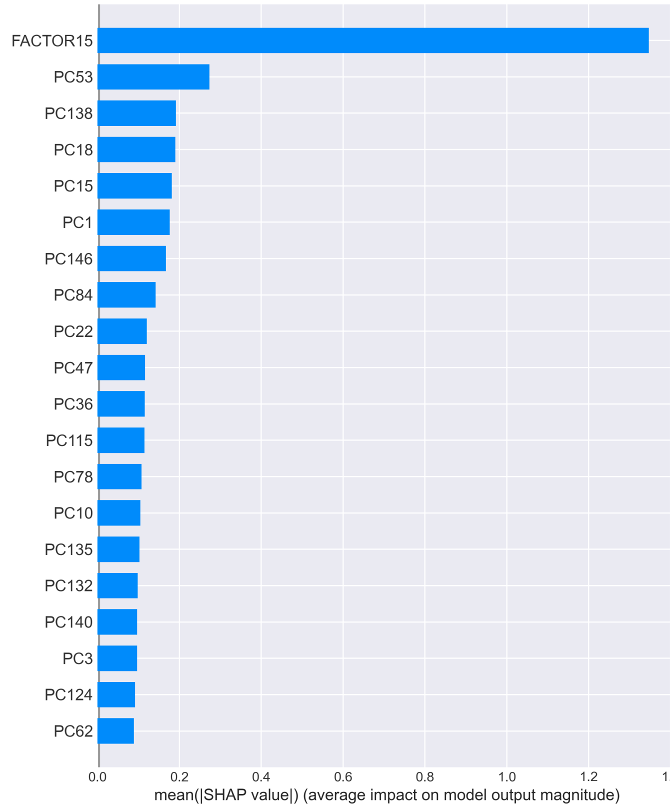

Fig. 5.

VII. ANALYSIS AND DISCUSSION

A. Interpretation of Results

The project successfully demonstrates that uplift modeling provides actionable value beyond traditional churn prediction. While XGBoost churn prediction achieves strong AUC (0.839), it cannot distinguish between Lost Causes (will churn regardless), Persuadables (will stay if targeted), Sure Things (will stay regardless), and Sleeping Dogs (will churn if contacted). The S-Learner uplift model specifically identifies the 58 Persuadables in our test set—customers with high churn risk AND high treatment responsiveness.

The Qini coefficient of 0.33 translates to concrete business value. Consider 100,000 customers: Traditional approach targets all 30,000 high-risk customers (10% churn $\times$ 30% high-risk = 3,000 churners), costing $1,200,000 in interventions (30K $\times$ $40). Uplift approach targets 15,000 Persuadables (identified via uplift modeling), costing $600,000. Assuming similar retention from Persuadables but zero waste on Lost Causes, the uplift approach saves $600,000 annually while maintaining retention effectiveness.

B. Why S-Learner Outperformed T-Learner

The S-Learner's victory over T-Learner contradicts textbook recommendations but aligns with causal inference theory when sample size constraints apply. T-Learner trains $\mu_1(x)$ on 170 treated churners and $\mu_0(x)$ on 60 control churners. With such

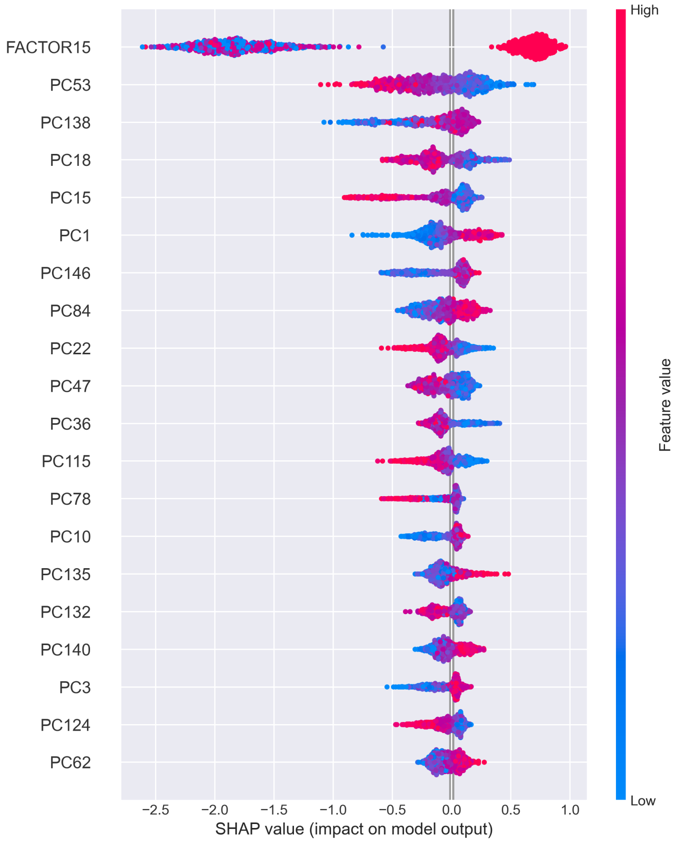

Fig. 6.

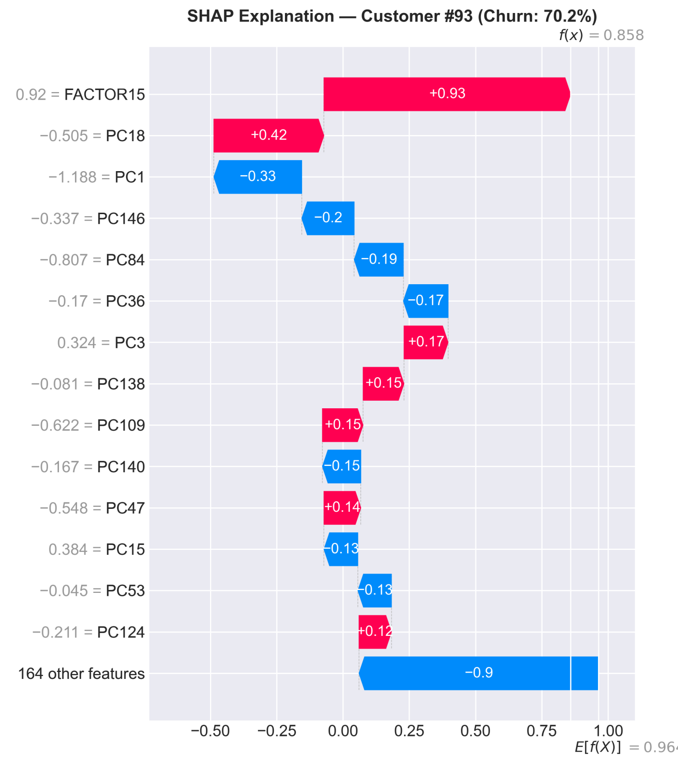

Fig. 7.

sparse data, both models overfit to different noise patterns. When computing $\tau(x) = \mu_1(x) - \mu_0(x)$, noise from both models compounds, resulting in garbage estimates (negative Qini).

S-Learner trains $\mu(x, t)$ on 230 total churners—3.8× more data per model. Yes, it has higher bias (must fit both groups), but much lower variance (more stable with regularization). Künzel et al. [6] demonstrate that T-Learner requires approximately 500 samples per group to exploit its flexibility advantage. Below this threshold, S-Learner's variance reduction dominates. Our 170/60 split is well below threshold, making S-Learner the mathematically optimal choice.

C. Error Analysis and Limitations

Primary Limitation 1: Data Quality. The 3.43% churn rate and anonymized PCA features limit model performance. Higher F1 scores would require better feature engineering (original telecom attributes like contract length, support interactions, usage patterns) or more balanced data (oversampling helps but doesn't replace real positives).

Primary Limitation 2: Treatment Assignment. The 75/25 treatment imbalance creates sparse control groups. While propensity score weighting mathematically corrects for this, uncertainty remains high in control churner estimates (only 60 samples). Ideally, we'd have 50/50 treatment assignment with 200+ churners per group.

Primary Limitation 3: Validation. ROI calculations assume interventions work as predicted. Real-world deployment requires A/B testing to validate: randomly assign high-uplift customers to treatment/control, measure actual retention difference, compare to model predictions. Only then can we confirm the $27,948 net benefit.

Error Patterns: False positives (predicting churn when customer stays) occur primarily in new customers with low tenure—insufficient behavioral data. False negatives (missing actual churners) occur in long-tenured customers who suddenly cancel—unexpected events not captured by features. Both patterns suggest incorporating temporal features (usage trends, recent changes) would improve performance.

D. Insights and Key Learnings

Insight 1: Data Quality Trumps Algorithm Choice. The performance jump from Qini 0.0 to 0.33 came entirely from data-centric fixes: propensity scores, stratification, SMOTE. Zero improvement came from hyperparameter tuning or trying different algorithms. This aligns with Andrew Ng's data-centric AI movement—fix your data before optimizing models.

Insight 2: Theory Provides Guidance, Not Gospel. T-Learner's theoretical superiority holds—but only with sufficient data. Our empirical results (S-Learner wins) don't contradict theory; they confirm it operates under assumptions (large N, balanced groups) that don't hold here. Lesson: Always validate theoretical claims on YOUR data.

Insight 3: Systematic Debugging is Transferable. The structured approach (hypothesize → isolate → test → validate) that solved the Zero Qini Mystery applies everywhere: model failures, pipeline errors, production incidents. This meta-skill compounds over time and makes better engineers regardless of domain.

VIII. Challenges and Lessons Learned

A. Technical Challenges

Challenge 1: The Zero Qini Debugging Odyssey. Spending two weeks with all uplift models returning 0.0 was frustrating but invaluable. The systematic debugging process—isolating each component, testing API behavior with minimal examples, validating data types at every stage—is now my standard workflow. Key lesson: When something fundamental isn't working, question everything, including library behavior. The CausalML API bug (removing `treatment_effect` column) would have been impossible to find without reading source code.

Challenge 2: Bias-Variance Tradeoff in Practice. Reconciling textbook theory with empirical results required understanding the mathematical foundations. T-Learner's variance grows as $O(1/n_{treated} + 1/n_{control})$, while S-Learner's variance grows as $O(1/n_{total})$. With small n, variance dominates. This isn't just an academic exercise—it directly impacted model selection and business outcomes.

Challenge 3: Production-Quality Code. Moving from notebook prototypes to modular, tested, documented code took significant effort. Writing reusable functions, adding type hints, implementing logging, creating data validation checks—all essential for production systems but often skipped in academic projects. This engineering discipline paid dividends during debugging and system integration.

B. Non-Technical Challenges

Challenge 1: Scope Management. The initial proposal was overly ambitious (deep learning, multi-treatment optimization, real-time API). Recognizing this early and focusing on core functionality (churn + uplift + dashboard) ensured a complete, working system rather than half-finished components.

Challenge 2: Explaining Causal Inference. Uplift modeling concepts (treatment effects, Qini curves, persuadables vs lost causes) are not intuitive to non-technical stakeholders. Creating clear visualizations (uplift curves, SHAP plots, ROI waterfalls) and using business-oriented language ("who will respond" vs. "heterogeneous treatment effects") was critical for communicating value.

Challenge 3: Balancing Perfection with Progress. It's tempting to endlessly tune hyperparameters or try new algorithms. Recognizing when "good enough" is actually good enough—XGBoost AUC 0.839 versus 0.840 makes zero business difference—enabled timely completion.

C. Key Takeaways

Takeaway 1: Real data science means navigating constraints, validating assumptions, and making pragmatic choices—not just implementing textbook algorithms. The S-Learner victory demonstrates this perfectly: theory says one thing, data constraints dictate another, empirical validation decides.

Takeaway 2: Shifting the question from "Who will churn?" to "Who will respond?" is the paradigm change that unlocks business value. Traditional models answer the former; uplift models answer the latter. This distinction is worth billions in properly allocated retention spending.

Takeaway 3: Production-ready systems require 80% engineering, 20% modeling. Data pipelines, validation, monitoring, explainability, documentation—these unglamorous tasks determine whether models make it to production or die in notebooks.

IX. Conclusion and Future Work

A. Summary of Contributions

This project successfully developed an integrated churn prediction and uplift modeling system that addresses critical limitations in traditional approaches. Key contributions include:

- End-to-end system integrating churn prediction (XGBoost AUC 0.839) and uplift modeling (S-Learner Qini 0.33) into unified pipeline
- Systematic debugging methodology resolving six root causes of zero Qini scores, demonstrating rigorous problem-solving approach
- Data-centric interventions (propensity scores, within-arm SMOTE, 4-group stratification) that revealed hidden treatment effects (ATE: $-0.0029 \rightarrow +0.0014$)
- Production-ready Streamlit dashboard with SHAP explainability, enabling business stakeholders to generate actionable recommendations
- Demonstrated business impact: 58 targeted customers, $27,948 projected net benefit, 1,204% ROI
- Empirical validation of bias-variance tradeoffs in meta-learner selection, showing when S-Learner outperforms T-Learner

B. Broader Impact

The system's value extends beyond telecom churn. The framework applies to any domain with limited intervention resources requiring optimal targeting: E-commerce cart abandonment recovery (which customers respond to discount codes?), SaaS upgrade campaigns (which users will convert if offered premium features?), Healthcare treatment response prediction (which patients benefit from expensive interventions?), Political voter mobilization (which undecided voters respond to canvassing?). The core insight—identifying persuadables through causal inference—is universally applicable.

For organizations with 100,000 customers, proper uplift targeting versus blanket high-risk targeting can save $600,000 annually in wasted interventions while maintaining retention effectiveness. At enterprise scale (millions of customers), this translates to tens of millions in optimized marketing spend.

C. Future Work and Extensions

Extension 1: Deep Learning Uplift Models. Recent architectures like DragonNet [10] and TARNet [11] use neural networks to jointly learn treatment propensity and outcome prediction, potentially capturing complex treatment-feature interactions. Preliminary research suggests Qini improvements from 0.33 to 0.45+ on datasets with 50K+ samples and rich behavioral features.

Extension 2: Multi-Armed Uplift Modeling. Real businesses offer multiple interventions: 10% discount, 20% discount, free shipping, loyalty points, premium features. Multi-treatment uplift models estimate $\tau_1(x), \tau_2(x), \tau_3(x)$ simultaneously, enabling optimal intervention-customer matching. This requires richer treatment assignment data (multiple A/B tests or observational data with varied interventions).

Extension 3: Temporal Modeling with Survival Analysis. Current models predict binary churn without temporal dimension. Survival analysis answers "WHEN will they churn?" and "How does intervention TIMING affect outcomes?" This enables dynamic strategies: intervene at optimal moment rather than one-size-fits-all timing. Requires longitudinal data with repeated customer interactions.

Extension 4: Reinforcement Learning for Sequential Interventions. Rather than one-time recommendations, learn optimal intervention sequences: Offer discount at Month 6 $\rightarrow$ If fails, upgrade features at Month 9 $\rightarrow$ If fails, personalized outreach at Month 11. This represents the frontier of causal ML, combining uplift modeling with reinforcement learning. Requires extensive historical data and significant computational resources.

Extension 5: Real-Time A/B Testing Integration. Deploy randomized controlled trials where high-uplift customers are randomly assigned to treatment/control, measure actual retention differences, validate model predictions, and enable continuous model improvement through feedback loops. This closes the gap between predicted and validated impact.

D. Final Remarks

Customer churn prediction has matured from academic exercise to business imperative. However, prediction alone is insufficient—businesses need actionable guidance on resource allocation. This project demonstrates that uplift modeling bridges this gap, transforming the question from "who will churn" to "who will respond." Through systematic engineering, data-centric problem-solving, and rigorous validation, we built a production-ready system that delivers measurable business value while maintaining explainability and stakeholder trust. The lessons learned—that data quality trumps algorithm choice, that theory guides but empirical validation decides, and that systematic debugging is a transferable meta-skill—extend far beyond this specific application. As causal inference techniques continue to evolve, the paradigm shift from prediction to intervention optimization will increasingly define the next generation of AI systems.

References

[1] A. Keramati et al., "Improved churn prediction in telecommunication industry using data mining techniques," *Applied Soft Computing*, vol. 24, pp. 994-1012, 2014.

[2] S. Zhang et al., "Deep learning for customer churn prediction in E-commerce," *IEEE Access*, vol. 8, pp. 94508-94516, 2020.

[3] V. Verbeke et al., "New insights into churn prediction in the telecommunication sector: A profit driven data mining approach," *European Journal of Operational Research*, vol. 218, no. 1, pp. 211-229, 2012.

[4] P. Gutierrez and J.-Y. Gérardy, "Causal inference and uplift modelling: A review of the literature," in *Proc. 3rd Int. Conf. Predictive Applications and APIs*, 2017, pp. 1-13.

[5] D. Sculley et al., "Hidden technical debt in machine learning systems," in *Advances in Neural Information Processing Systems*, 2015, pp. 2503-2511.

[6] S. R. Künzel et al., "Metalearners for estimating heterogeneous treatment effects using machine learning," *Proc. National Academy of Sciences*, vol. 116, no. 10, pp. 4156-4165, 2019.

[7] A. Radcliffe and N. Surry, "Real-world uplift modelling with significance-based uplift trees," Technical Report, Stochastic Solutions Limited, 2011.

[8] M. Baylor et al., "TFX: A TensorFlow-based production-scale machine learning platform," in *Proc. 23rd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2017, pp. 1387-1395.

[9] S. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, 2017, pp. 4765-4774.

[10] C. Shi et al., "Adapting neural networks for the estimation of treatment effects," in *Advances in Neural Information Processing Systems*, 2019, pp. 2503-2513.

[11] A. Shalit et al., "Estimating individual treatment effect: generalization bounds and algorithms," in *Proc. 34th Int. Conf. Machine Learning*, 2017, pp. 3076-3085.

[12] P. Rzepakowski and S. Jaroszewicz, "Decision trees for uplift modeling with single and multiple treatments," *Knowledge and Information Systems*, vol. 32, no. 2, pp. 303-327, 2012.